

BSDS MATHEMATICS II FINAL EXAMINATION FIRST PART

Date: 18th May, 2026

Time: 90 minutes

Total Marks: 25

- (1) ($3 + 3 + 3 + 4 = 13$ marks) Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}. \text{ Find a diagonal matrix } D \text{ and an orthogonal matrix } S \text{ such that } S^{-1}AS = D.$$

- (2) (4 marks) Let $f(x, y) = (\cos(x), e^y, x^2 + y^2)$ and $g(x, y, z) = (\sin(z), x + y^4)$. Using the chain rule for derivatives of vector fields, find $D(g \circ f)(0, 0)$.
- (3) ($4 + 4 = 8$ marks) Compute the double integral $\iint_S \sqrt{1 - x^2 - y^2} dx dy$, where $S = \{(x, y) | x^2 + y^2 \leq 1, x \geq 0, y \geq 0\}$ in two ways, directly and using polar coordinates.

BSDS MATHEMATICS II FINAL EXAMINATION SECOND PART

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Total Marks: 25

- (1) (3 + 4 + 5 = 12 marks) Consider the map $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(x, y) = x^2 - xy + y^2 - 2x + y$. Find the directional derivative of f at $(1, 1)$ along the direction $(2, 3)$. Find all the stationary points of f . Classify these stationary points as relative maxima or relative minima or saddle point. 4/5
- (2) (3 + 3 = 6 marks) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be $f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}$ if $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$. Compute $\frac{\partial^2 f}{\partial x \partial y}(0, 0)$ and $\frac{\partial^2 f}{\partial y \partial x}(0, 0)$. 4/3
- (3) (3 marks) Find the radius of convergence for the power series $\sum_{n=1}^{\infty} (\frac{3^n}{n4^n}) x^n$. 0
- (4) (4 marks) For $x \in [0, \infty)$, let $f_n(x) = \frac{x}{n}$, find the pointwise limit function f of the sequence (f_n) and determine whether $f_n \rightarrow f$ uniformly on $[0, \infty)$. No